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Reg No.: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree 7th semester (S,FE) Exam April 2025 (2019 Scheme)

Course Code: EET413

Course Name: ELECTRIC DRIVES

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

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| 1 | Derive the dynamic torque equation of a motor load system. | (3) |
| 2 | Explain the different components of load torque. | (3) |
| 3 | From necessary waveforms, derive the equations of rms value of source current and power factor of a single-phase full converter drive. | (3) |
| 4 | A separately excited dc motor, operating from a single-phase half-controlled bridge at a speed of 1400rpm, has an input voltage of $330 \sin 314t$ and a back emf 80V. The SCRs are fired symmetrically at $\alpha=30^\circ$ in every half cycle and the armature has a resistance of 4Ω . Calculate the average armature current and the motor torque. | (3) |
| 5 | With suitable circuit diagram and waveforms, explain the motoring control operation of DC Chopper drives. | (3) |
| 6 | Draw the circuit diagram of a class-C chopper fed DC motor. Also draw its V/I characteristics. | (3) |
| 7 | Explain the speed control method of induction motor with stator voltage and also state the disadvantages of this method | (3) |
| 8 | List out any three methods employed for the speed control of Induction Motor. | (3) |
| 9 | With block diagram, explain the variable frequency control of SM drive in self-control mode. | (3) |
| 10 | What are the advantages of Space Vector PWM? | (3) |

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

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| 11 | a) What are the different classifications of electric drive? Give examples for each. | (4) |
| | b) Explain in detail about four quadrant operation of a hoist load. | (10) |

OR

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| 12 | a) Derive the mathematical expression for steady state stability analysis of equilibrium operating point in electrical drive. | (6) |
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- b) Derive the expressions to find the equivalent values of load torque and moment of inertia of the Electric drive system in Rotational motion. (8)

Module II

- 13 a) Explain the working of 3-phase fully-controlled separately excited DC drive with necessary waveforms. (8)
- b) A 220V, 1500rpm, 10A separately excited dc motor has an armature resistance of 1Ω . It is fed from a single phase fully controlled bridge rectifier with an ac source voltage of 230V, 50Hz. Assuming continuous load current, compute: (6)
- (i) Motor speed at the firing angle of 30° and torque of 5Nm
- (ii) Developed torque at the firing angle of 45° and speed of 1000rpm.

OR

- 14 a) Explain the working of a dual converter (circulating current type) fed separately excited DC motor. (6)
- b) Explain the operation of a single phase fully controlled rectifier fed separately excited DC motor with necessary waveforms. (8)

Module III

- 15 a) Explain the operation of four quadrant chopper fed DC drives. (10)
- b) The chopper used for on-off control of a dc separately excited motor has supply voltage of 230V dc, an on time of 10msec and off-time of 15msec. Neglecting armature inductance and assuming continuous conduction of motor current, calculate the average load current when the motor speed is 1500rpm and has a motor constant of $K_m=0.5\text{V/rad per sec}$. The armature resistance is 3Ω . (4)

OR

- 16 a) Explain the regenerative braking control operation of a separately excited dc motor. Also derive the equation of motor speed during regenerative braking using chopper-fed DC drives. (7)
- b) A dc chopper is used for regenerative braking of a separately excited dc motor. The dc supply voltage is 400V. The motor has $R_a=0.2\Omega$, $K_m=1.2\text{Vsec/rad}$. The average armature current during regenerative braking is kept constant at 300A with negligible ripple. For a duty cycle of 60% for a chopper, determine: (7)
- (i) Power returned to the dc supply
- (ii) Equivalent load resistance of motor acting as a generator
- (iii) Minimum and maximum permissible braking speeds
- (iv) Speed during regenerative braking

Module IV

- 17 a) Explain the static rotor resistance control method for the speed control of a slip ring induction motor. What are the disadvantages of this method? (7)
- b) Explain in detail, the v/f control applied to induction motor drives. (7)

OR

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- 18 a) Explain the static Kramer scheme for the speed control of a slip ring Induction motor. (7)
- b) With neat block diagram, explain the closed loop speed control of VSI fed induction motor drives. (7)

Module V

- 19 a) With neat diagrams, explain the concept of Space Vector Modulation. (10)
- b) Explain the Clarke transformation. (4)

OR

- 20 a) Explain the field-oriented control (FOC) of an AC motor with a block diagram. (7)
- b) With neat block diagram, explain the open loop v/f control of VSI fed synchronous motor drives. What is the need of delay circuit in open loop v/f control? (7)
